

Alexander Terry

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PROFESSIONAL SUMMARY

Embedded Software Engineer with hands-on experience developing C/C++ applications for custom telecom hardware and real-time Linux environments. Proficient in building and configuring custom Linux distributions using Yocto and Buildroot, integrating SIP, RTP, TCP/UDP protocol stacks, and implementing audio codecs for low-latency communication. Skilled in low-level system optimization, cross-compilation for embedded architectures, and automating build workflows using Bash and Perl.

EDUCATION

The State University of New York at Buffalo

Aug 2021 - May 2025

B.S. Computer Science & B.A. Mathematics | Honors College | GPA: 3.92 / 4.00

Certifications: Google Cybersecurity, IBM Z Xplore Concepts, CITI Biomedical Research Investigators.

WORK EXPERIENCE

Software Engineer | General Dynamics Mission Systems | Scottsdale, AZ

Sep 2025 - Present

- Identified and proposed an open-source alternative to a planned commercial solution, projected to save the program over \$100,000 pending adoption.
- Develop embedded C/C++ software across the full lifecycle, from driver/firmware development through build system tooling, working with cross-compilation toolchains, RTOS/RTEMS environments, and FPGA-integrated hardware within a security controlled defense engineering environment.
- Build and maintain containerized development and build environments, plus Python unit and integration test suites, to standardize embedded builds and support regression testing across build targets.
- Leverage AI coding assistants (ChatGPT, Claude) throughout the development workflow to accelerate implementation, debugging, and code review.

Software Engineer | VOCAL Technologies | Buffalo, NY

Jan 2025 - July 2025

- Developed and optimized embedded C software for real-time voice and data transmission systems (MoIP, VoIP, FoIP), debugging and tuning network protocol stacks (SIP, RTP, TCP/UDP) and audio codecs (G.711, G.729) to meet latency and bandwidth constraints.
- Built and optimized custom Linux distributions using Yocto and Buildroot for ARM-based embedded hardware, including cross-compilation toolchains and low-level performance tuning for real-time applications.
- Automated build and deployment workflows with Perl and Bash, reducing cycle times and improving CI/CD reliability.
- Implemented and optimized signal modulation and compression techniques at various bitrates to enhance data transmission efficiency and reliability in embedded telecom systems.

PROJECTS

BME280 Pi Telemetry | Sensor telemetry pipeline on a Raspberry Pi 5 | [bme280-pi-telemetry](#)

C, Python, FastAPI, PostgreSQL, Docker, JavaScript

- Wrote a C driver to read the BME280 over SPI, including calibration and compensation, then built a Python/FastAPI service on top that stores readings in PostgreSQL and serves them through a REST API, with a small JavaScript dashboard for live and historical data plus CSV export.
- Handled reconnecting the sensor after failed reads, rejecting out-of-range data, and pruning old readings automatically, and packaged the service with Docker and GitHub Actions CI for ARM64/AMD64 builds.

Jaclyn's Threading Website | Full stack booking platform for a local beauty salon | [jaclynbeauty.com](#)

Node.js, Express, PostgreSQL, Docker, JavaScript, HTML/CSS

- Architected and shipped a production booking system end-to-end as sole engineer, handling real-time scheduling, automated notifications, and an authenticated admin dashboard for a live business.
- Hardened the platform with enforced CSP, rate limiting, bcrypt auth, and layered bot mitigation, and eliminated a live double-booking defect via a zero-downtime database migration.

TECHNICAL SKILLS

Programming Languages: C, C++, Python; familiar with Rust.

Embedded Build & Debug Tools: Yocto, Buildroot, U-Boot, Device Tree, Linux Kernel Modules/Drivers, GCC/Clang Cross-Toolchains, Make, CMake, GDB, Valgrind.

Networking & Protocols: SIP, RTP, TCP/IP, UDP, VoIP, FoIP.

Hardware & Interfaces: ARM (Cortex-A/M series), MIPS, UART, I2C, SPI, JTAG, HSDP.

Dev & Lab Tools: Git, GitLab CI, Jira, Docker, Wireshark, Signal Generators, Spectrum Analyzers.